

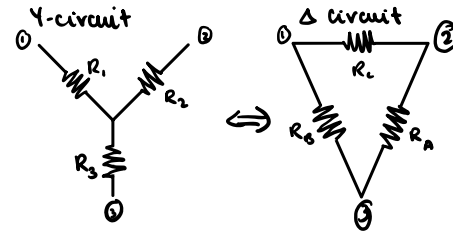
Charges and Current

$$I = \frac{dq}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta q}{\Delta t} = \frac{Q}{\tau} \quad \tau = \frac{L}{v}$$

$$\frac{Q}{\tau} = \frac{Qv}{L} = \frac{Q_v A L v}{L} \quad Q_v: \text{charge density}$$

$$I = \frac{Q}{\tau} = Q_v A v \quad q(t) = \int_0^t i(t') dt'$$

Delta-Y Transforms



$Y \rightarrow \Delta$

$$R_A = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_B = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

$$R_C = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

$\Delta \rightarrow Y$

$$R_1 = \frac{R_B R_C}{R_A + R_B + R_C}$$

$$R_2 = \frac{R_A R_C}{R_A + R_B + R_C}$$

$$R_3 = \frac{R_A R_B}{R_A + R_B + R_C}$$

Source Transformations

Voltage

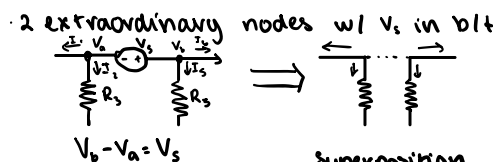
$$V_{\text{high}} - V_{\text{low}} = \frac{W}{q}$$

Voltage diff. same regardless of ground point

Power

$$P = \frac{dW}{dt} = \frac{dW}{dq} \cdot \frac{dq}{dt} = V \cdot i$$

Supernode



Waves

Sinusoidal signals (time invariant): $\sin(kx)$

Sinusoidal signal (@ single location): $\sin(\omega t)$

Traveling wave: $\sin(kx - \omega t)$ or $\sin(kx + \omega t)$

$$k: \text{wavevector} = \frac{2\pi}{\lambda} \quad \lambda: \text{wavelength}$$

$$\omega: \text{angular frequency} = \frac{2\pi}{T} \quad T: \text{period}$$

$$v: \text{phase vel. (f}\lambda) \quad \omega = 2\pi f$$

Mobility & Resistivity

m: mass of charge

$$\text{Mobility: } \mu = \frac{q \tau_{\text{avg}}}{m}$$

τ_{avg} : avg. time b/t 2 events

n: electron density

$$J = \frac{I}{A} = Q_v v = nq \mu E = \sigma E$$

Q_v : charge volume density

Equivalent Resistances

Series: $I_{\text{tot}} = I_1 = I_2 = \dots$ Parallel: $V_{\text{tot}} = V_1 = V_2 = \dots$

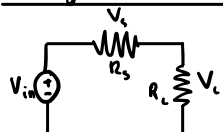
$$V_{\text{tot}} = V_1 + V_2 + \dots$$

$$I_{\text{tot}} = I_1 + I_2 + \dots$$

$$R_{\text{tot}} = R_1 + R_2 + \dots$$

$$\frac{1}{R_{\text{tot}}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

Voltage/Current Divider (Resistors)

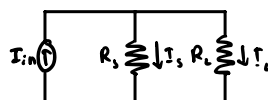


Voltage Divider

$$I_{\text{tot}} = \frac{V_{\text{in}}}{R_{\text{tot}}} = \frac{V_{\text{in}}}{R_1 + R_2}$$

$$V_1 = \frac{V_{\text{in}} R_1}{R_1 + R_2}$$

$$V_2 = \frac{V_{\text{in}} R_2}{R_1 + R_2}$$



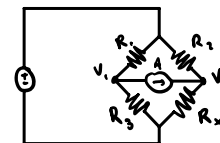
Current Divider

$$V_{\text{tot}} = I_{\text{in}} R_{\text{tot}} = \frac{I_{\text{in}} R_1 R_2}{R_1 + R_2}$$

$$I_1 = \frac{V_{\text{tot}}}{R_1} = \frac{I_{\text{in}} R_2}{R_1 + R_2}$$

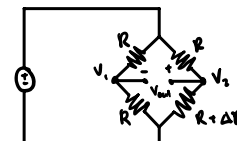
$$I_2 = \frac{V_{\text{tot}}}{R_2} = \frac{I_{\text{in}} R_1}{R_1 + R_2}$$

Wheatstone Bridge



Balanced condition:

$$R_x = \left(\frac{R_1}{R_2}\right) R_3$$



SENSOR:

$$V_{\text{out}} = \frac{V_0}{4} \left(\frac{\Delta R}{R} \right)$$

Analysis By Inspection

Mesh Current Analysis

$$\begin{bmatrix} R_{11} & R_{12} & \dots & R_{1n} \\ R_{21} & R_{22} & \dots & R_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ R_{n1} & R_{n2} & \dots & R_{nn} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_n \end{bmatrix} = \begin{bmatrix} V_{s1} \\ V_{s2} \\ \vdots \\ V_{sn} \end{bmatrix}$$

R_{kk} = sum of all resistances in mesh k

R_{kl} = neg. sum of resistances shared by k and l (0 if none)

I_k = current of mesh k

V_k = total of ind. V_s in k (+ for voltage rise in $\rightarrow$ direction)

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Nodal Analysis

$$\begin{bmatrix} G_{11} & G_{12} & \dots & G_{1n} \\ G_{21} & G_{22} & \dots & G_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ G_{n1} & G_{n2} & \dots & G_{nn} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_n \end{bmatrix} = \begin{bmatrix} I_{s1} \\ I_{s2} \\ \vdots \\ I_{sn} \end{bmatrix}$$

G_{kk} = sum of all conductances for node k

G_{kl} = neg. sum of conductances connecting node k and l (0 if none)

V_k = voltage at node k

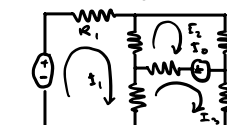
I_k = total of I_s entering node k

(neg. for current leaving)

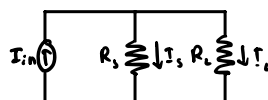
Supermesh

2 meshes sharing same

current source



$$I_0 = I_2 - I_3$$



Current Divider

$$V_{\text{tot}} = I_{\text{in}} R_{\text{tot}} = \frac{I_{\text{in}} R_1 R_2}{R_1 + R_2}$$

$$I_1 = \frac{V_{\text{tot}}}{R_1} = \frac{I_{\text{in}} R_2}{R_1 + R_2}$$

$$I_2 = \frac{V_{\text{tot}}}{R_2} = \frac{I_{\text{in}} R_1}{R_1 + R_2}$$

2nd Order Circuits

Initial & Boundary Conditions:

- i_L cannot change instantaneously
- V_C cannot change instantaneously
- $i_L(t=0^-) = i_L(t=0)$ & $V_C(t=0^-) = V_C(t=0)$

Steps:

- draw out circuits @ $t=0^-$, $t=0$, $t=\infty$
- inductors: initially act as open circuit
→ eventually short circuit
- capacitors: initially act as short circuit
→ eventually open circuit

write KVL / KCL's:

$$V_L = L \frac{di_L}{dt} \quad i_C = C \frac{dV_C}{dt} \quad (\text{can also take } \frac{d}{dt})$$

$$x'' + ax' + bx = c \quad 2\alpha = a \quad \omega_0^2 = b \quad \omega_0 = \sqrt{b}$$

$$\text{IC: } x(0) \quad x'(0) \quad \text{FC: } x(\infty) = \frac{c}{b}$$

Overdamped ($\alpha > \omega_0$)

$$x(t) = [A_1 e^{s_1 t} + A_2 e^{s_2 t} + x(\infty)] u(t)$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} \quad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

$$A_1 = \frac{x'(0) - s_2 [x(0) - x(\infty)]}{s_1 - s_2}$$

$$A_2 = -\frac{[x'(0) - s_1 [x(0) - x(\infty)]]}{s_1 - s_2}$$

Critically Damped ($\alpha = \omega_0$)

$$x(t) = [(B_1 + B_2 t) e^{-\alpha t} + x(\infty)] u(t)$$

$$B_1 = x(0) - x(\infty) \quad B_2 = x'(0) + \alpha [x(0) - x(\infty)]$$

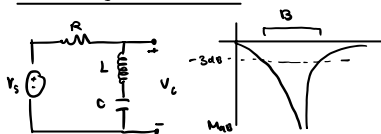
Underdamped ($\alpha < \omega_0$)

$$x(t) = [(D_1 \cos \omega_d t + D_2 \sin \omega_d t) e^{-\alpha t} + x(\infty)] u(t)$$

$$D_1 = x(0) - x(\infty) \quad D_2 = \frac{x'(0) + \alpha [x(0) - x(\infty)]}{\omega_d}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

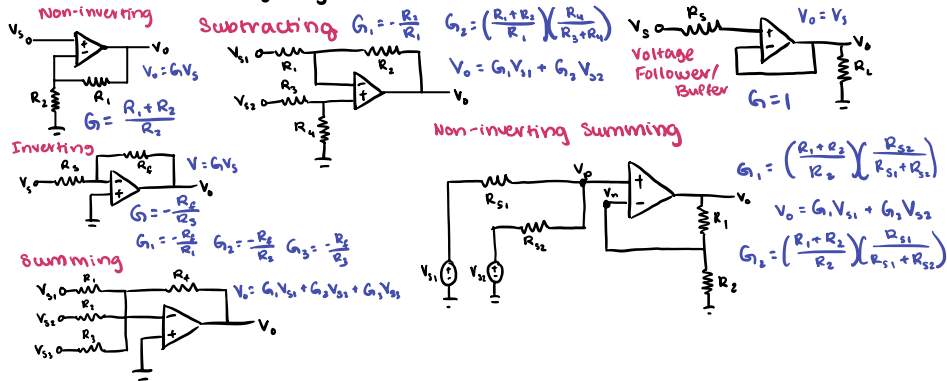
Bandreject Filter



Op Amps

Ideal: $I_- = I_+ = 0$

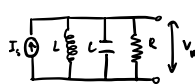
$V_- = V_+$ (only neg. feedback)



Bandpass Filter



Parallel RLC



$$H(\omega) = \frac{V_o}{V_s} \quad \omega_0 = \frac{1}{\sqrt{LC}} \quad B = \frac{R}{L}$$

$$Q = \frac{\omega_0}{B} = \frac{\omega_0 L}{R} \quad B = \omega_{c2} - \omega_{c1} \quad \omega_0 = \sqrt{\omega_{c1} \omega_{c2}}$$

$$\omega_{c1} = \left[-\frac{1}{2Q} + \sqrt{1 + \frac{1}{4Q^2}} \right] \omega_0 \quad \omega_{c2} = \left[\frac{1}{2Q} + \sqrt{1 + \frac{1}{4Q^2}} \right] \omega_0$$

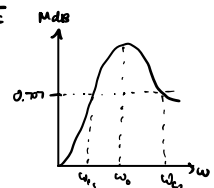
$$\omega_{c2} = \left[\frac{1}{2Q} + \sqrt{1 + \frac{1}{4Q^2}} \right] \omega_0 \quad \omega_0 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$H(\omega) = \frac{V_o}{I_s} \quad \omega_0 = \frac{1}{\sqrt{LC}} \quad B = \frac{1}{RC}$$

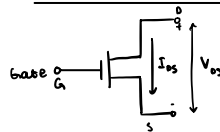
$$Q = \frac{\omega_0}{B} = \frac{R}{\omega_0 L}$$

$$\omega_{c1} = \left[-\frac{1}{2Q} + \sqrt{1 + \frac{1}{4Q^2}} \right] \omega_0$$

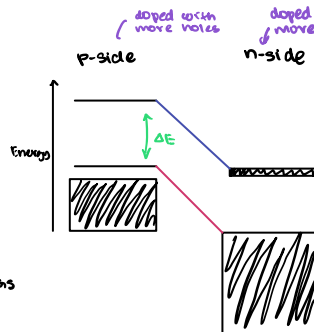
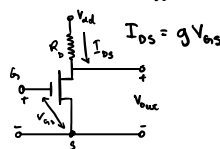
$$\omega_{c2} = \left[\frac{1}{2Q} + \sqrt{1 + \frac{1}{4Q^2}} \right] \omega_0$$



MOSFET / Devices

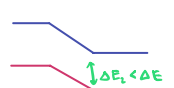


$$V_{out} = V_{DD} - I_{DS} R_D$$

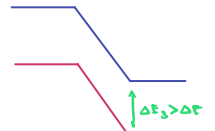


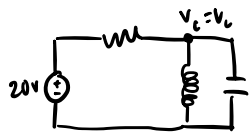
Forward Bias:

$V_0 > 0$



Reverse Bias





$$\frac{V_L - 20V}{200} + i_L + i_C = 0$$

$$\frac{V_L}{200} + i_L + i_C = \frac{1}{10}$$

$$V_L = L \frac{di_L}{dt} \quad \frac{dV_L}{dt} = \frac{V_L}{L} = \frac{V_C}{L}$$

$$\frac{1}{200} \frac{dV_L}{dt} + \frac{di_C}{dt} + \frac{di_L}{dt} = 0$$

$$i_C = C \frac{dV_C}{dt} \quad \frac{di_C}{dt} = C \frac{d^2 V_C}{dt^2}$$

$$\frac{1}{200} \frac{dV_C}{dt} + C \frac{d^2 V_C}{dt^2} + \frac{V_C}{L} = 0$$

$$\frac{d^2 V_C}{dt^2} + \frac{1}{200L} \frac{dV_C}{dt} + \frac{1}{LC} V_C = 0$$

$$\frac{d^2 V_C}{dt^2} + 2000 \frac{dV_C}{dt} + 160000 V_C = 0$$

$$2\alpha = 2000 \quad \alpha = 1000 \quad \omega_0^2 = 160000 \quad \omega_0 = 400$$

$\alpha < \omega_0$ under damped



$$i_L + i_C = 0$$

$$V_L = L \frac{di_L}{dt}$$

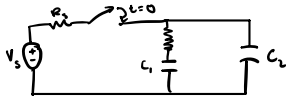
$$i_C = C \frac{dV_C}{dt}$$

$$\frac{di_L}{dt} + \frac{di_C}{dt} = 0$$

$$\frac{dV_C}{dt} = \frac{V_C}{L}$$

$$\frac{di_C}{dt} = C \frac{d^2 V_C}{dt^2}$$

$$\frac{d^2 V_C}{dt^2} + \frac{1}{LC} V_C = 0$$



$t = 0^-$:

$$V_{C_1}(0^-) = V_{C_2}(0^-) = 0V$$

$$V_{C_1}(0^-) = V_{C_2}(0^-) = 0V$$

$t = 0$:

$i_{C_1}(0) = 0A \Rightarrow$ all current goes through C_2 w/c shorted

$$i_{C_2}(0) = \frac{V_s}{R_s} = \frac{10}{10^3} = 100 \mu A$$

$$V_{C_1}'(0) = \frac{i_{C_1}(0)}{C_1} = 0$$

$$\frac{V_{C_1} + R_s V_{C_1}' - V_s}{R_s} + C_1 V_{C_1}'' + C_2 [V_{C_1}' + R_s V_{C_1}''] = 0$$

$$V_{C_2}'(0) = \frac{i_{C_2}(0)}{C_2} = \frac{10^{-4}}{2 \cdot 10^{-6}} = 50 V/s$$

$$R_s C_2 V_{C_1}'' + \left(\frac{R_s C_1}{R_s} + C_1 + C_2 \right) V_{C_1}' + \frac{V_{C_1}}{R_s} - \frac{V_s}{R_s} = 0$$

$$V_{C_1}'' + \left(\frac{R_s C_1 + R_s C_2 + R_s^2 C_2}{R_s R_s C_1 C_2} \right) V_{C_1}' + \frac{1}{R_s R_s C_1 C_2} V_{C_1} = \frac{V_s}{R_s}$$

At $t > 0$:

$$V_{C_1}'' + a V_{C_1}' + b = c$$

$$\frac{V_{C_1} - V_s}{R_s} + i_{C_1} + i_{C_2} = 0$$

$$a = 6.5 \quad \alpha = \frac{a}{2} = 3.25$$

$$b = 5 \quad \omega_0^2 = b \Rightarrow \omega_0 = \sqrt{5}$$

$$c = 50 \quad = 2.24$$

$\alpha > \omega_0$:

$$i_{C_1} = C_1 V_{C_1}' \quad i_{C_2} = C_2 V_{C_2}'$$

$$V_{C_2}(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + V_{C_2}(\infty)$$

$$V_{C_2} = V_{C_1} + R_s i_{C_1} = V_{C_1} + R_s C_1 V_{C_1}'$$

$$V_{C_1}(\infty) = V_{C_2}(\infty) = V_s = 10V$$

$$V_{C_1}' = V_{C_1}' + R_s C_1 V_{C_1}''$$